Rithvik Vootukur

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Education

Texas A&M University College Station, TX
Bachelor of Science in Aerospace Engineering | Engineering Honors Program | GPA: 3.70/4.00 *Expected May 2027*

- **Relevant Coursework:** Aerodynamics, Aerospace Structures, Spacecraft Design, Propulsion, Numerical Methods
- **Memberships:** American Institute of Aeronautics and Astronautics (AIAA); ASME

Experience

Research Assistant at Turbulence and Advanced Computations Laboratory Jun 2026 – Present
TACL, Texas A&M University *College Station, TX*

- Analyze RIPTIDE simulation outputs to study how filtering turbulence scales just above the Kolmogorov microscale affects laser beam wander and scintillation.
- Verified and visualized a 4,218-case RIPTIDE sweep (111 focal-length and propagation-distance pairs, each with 37 filter scales from 0.1 mm to 3 m) in Python and Matplotlib on Linux.

Team Lead at TAMU-SPIRIT-2 Aug 2026 – Present
Senior Design Capstone (AERO 402), Texas A&M University *College Station, TX*

- Lead a 16-person team in a Texas A&M and Aegis Aerospace program preparing four experiments for an external International Space Station (ISS) platform based on the MISSE architecture.
- Edited and presented the System Definition Review (SDR), covering mission goals, requirements traced to goals with verification methods, concept of operations, and the plan to Preliminary Design Review (PDR).
- Contribute hands-on to the structures and electronics subsystems: run Abaqus finite element analysis (FEA) for structural analysis; conduct battery research and trade studies for thermal and power planning.

Project Manager and Team Lead, NASA Gateways to Blue Skies Competition 2023 – 2024
Texas A&M University *College Station, TX*

- Led a six-member multidisciplinary team to submit a 14-page technical proposal for a conceptual spherical search-and-rescue unmanned aerial vehicle (UAV); managed subsystem work and the project schedule.
- Originated the roll-cage concept and developed the SolidWorks CAD model, geometry sizing, and weight estimates; supported materials selection with Pugh-chart trade studies.
- Wrote nanomaterials documentation and delivered an approximately 20-minute technical presentation with Q&A.

Projects

AS 35 Mi Sailplane Wing Design | *SolidWorks, XFOIL, CAD, Airfoil Trade Study*

- Built a parametric SolidWorks model of the wing and surrounding aircraft geometry to support design iteration.
- Compared 10 glider airfoil candidates on aerodynamic performance metrics and documented the trade study for design selection.

RS-25 Nozzle and Compressible Flow Solver | *Python, Numerical Methods*

- Developed a Python solver for one-dimensional supersonic nozzle flow using the area-Mach relation and bracketing root finding to compute Mach number, pressure, temperature, density, and velocity from throat to exit.
- Vacuum thrust came within 5.1% of public RS-25 data; traced Isp and mass-flow gaps to the assumed gas constant.

Heat Transfer Numerical Solvers | *Finite Difference Method, Finite Element Method (FEM)*

- Implemented 2nd- and 4th-order finite difference and 1st- and 2nd-order finite element models for 1D steady conduction with material interfaces; Richardson extrapolation confirmed the expected convergence orders.

Technical Skills

Programming: Python, MATLAB, C++, Fortran, NumPy, SciPy, Matplotlib
Engineering Software: SolidWorks, Abaqus, XFOIL
Analysis Methods: Computer-aided design (CAD), finite element analysis (FEA), finite difference methods, compressible flow, trade studies
Data and Computing: HDF5, Linux, SVN version control, data analysis and visualization